MILITARY SPECIFICATION

Silica Gel

(For use in ammunition)

1. SCOPE

1.1 This Specification covers one type of silica gel for use in Aerial Mines (see 6.4).

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

RR-S-366	- Sieve, Test
TT-E-485	- Enamel, Semi-gloss, Rust Inhibiting

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedure and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-1235	- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes
MS 20003	- Indicator, Humidity, Card
MS 26507	- Indicator Card, Desiccant

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

FSC 1345

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2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Damgerous Articles

(The Interstate Commerce Commission Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, US Government Printing Office, Washington, D.C. 20402. Orders for the above publication should cite "49 CFR 71-78 (latest revision)".)

Officical Classification Committee Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33d Street, New York, New York 10016.)

3. REQUIREMENTS

3.1 Material.-The material shall conform to the requirements of Table I.

TABLE I

Characteristic	Requirement	Applicable Paragraph
Silica, %	99.6, min	4.3.1
Iron, as Fe_2O_3, %	0.03, max	4.3.2
pH	3.5, min	4.3.3
Density, lbs per cu. ft.	45-48	4.3.4
Water vapor absorption capacity at 77°F., %		4.3.5
At 10% relative humidity	5.7 min	
At 20% relative humidity	10.5 min	
At 40% relative humidity	22.0 min	
At 60% relative humidity	33.25 min	
At 80% relative humidity	36.0 min	
Granulation		4.3.6
Retained on #6, %	2.0 max	
Thru #6, retained on #10, %	55.0 min	

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Thru #10, retained on #16, %	19.0 min
Thru #16, %	3.5 max
Thru #20, %	0.5 max

3.2 Workmanship.-The material shall be free of dirt, chips and foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein.

4.1.1 Contractor quality assurance system.-If the contractor desires to use a quality assurance system which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the Contracting Officer for approval prior to start of production. It shall include a description covering controls for lot formation and identification, inspections to be performed, inspection stations, sampling procedures, methods of inspection (measuring and testing equipment), and provisions for control and disposition of non-conforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by the provisions of 4.2 and 4.3 and other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification, provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedures of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure which affect the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission or product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

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a. A statement that the lot complies with all requirements and provisions of the approved current written description of the material.

b. Specification number and date, together with an identification and date of changes.

c. Certificates of analysis of all material used directly by the contractor when such material is controlled by Government specification shall be made available upon request by the Contracting Officer.

d. Quantity of product in the lot.

e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 First article inspection.-

4.1.3.1 Submission.-The contractor shall submit a first article quantity as designated by the Contracting Officer for evaluation in accordance with the provisions of 4.3.3.2 (see 6.1). The first article sample shall consist of ten (10) pounds of material. The sample submitted shall have been produced by the contractor (or furnished by a supplier) using the same production processes, procedures, and equipment as will be used in fulfilling the contract. All parts and materials, including packaging and packing, shall be obtained from the same sources of supply as will be used in regular production. The sample shall be accompanied by certificates of conformance. A first article quantity or portion thereof, as directed by the Contracting Officer, shall also be submitted whenever there is a lapse in production for a period in excess of 90 days, or whenever a change occurs in manufacturing process, material used, drawing, specification or source of supply as to significantly affect product uniformity as determined by the Government. Prior to submission, the contractor shall inspect the sample to the degree necessary to assure that it conforms to the requirements of the contract and submit a record of this inspection with the sample. A sample containing known defects will not be submitted unless specifically authorized by the Contracting Officer.

4.1.3.2 Inspections to be performed.-The sample will be subjected by the Government to any or all of the examinations or tests specified in 4.2 and 4.3 of this specification and any or all requirements of the applicable drawings.

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4.1.3.3 Rejection.-If the sample fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of a sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless otherwise directed by the Contracting Officer.

4.2 Inspection provisions.-

4.2.1 Lot formation.-A lot shall consist of one or more batches of silica gel produced by one manufacturer in accordance with the same specification, or the same specification revision, under one continuous set of operating conditions. Each batch shall consist of that quantity of silica gel that has been subjected to the same unit chemical or physical mixing process intended to make the final product homogeneous. The product shall be submitted for inspection in accordance with MIL-STD-105 or MIL-STD-1235, as applicable.

4.2.2 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with MIL-STD-105. Continuous sampling plans, in accordance with MIL-STD-1235, may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect.

4.2.2.1 Drums

Categories	Defects	Method of Inspection	Code No. (See 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Weight of contents	Scale	01001
102.	Closure incomplete or damaged	Visual	01002
103.	Drum damaged	Visual	01003
Minor:	AQL 0.65 percent		
201.	Marking misleading or unidentifiable	Visual	01004

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4.2.3 Sampling.-A random sample of drums shall be selected from each lot in accordance with Level I of MIL-STD-105. A quantity of 5 to 10 pounds shall be accumulated by taking approximately equal portions from the sample drums and transferring quickly into a clean, dry container, which when sealed shall be air- and moisture-tight. This sample shall be thoroughly mixed but care should be taken not to mix so vigorously that particle size is significantly affected. This sample shall be tested for conformance to the requirements of 3.1 as specified in 4.3. If the sample fails to comply with any of the requirements specified in 3.1, the lot shall be rejected.

4.3 Testing.-

4.3.1 Silica - Major Defect - Defect Code 02001.-Heat about five (5) grams of material overnight in an oven at 350°F. Cool in a desiccator, then weigh accurately about one (1) gram of dried material. Transfer to a clean platinum dish of about 100 ml capacity. Add approximately 20 ml of 1:1 hydrochloric acid and evaporate to dryness. Cool the dish and cover the residue with 10% sulfuric acid. In a fume hood add slowly 20 ml of 1:1 hydrofluoric acid. After the vigorous action has subsided, evaporate and ignite to drive off the acid. Cool and weigh.

Calculations:

$$\text{Silica } \% = \frac{(A - B)\ 100}{A}$$

where A = weight of dried sample
B = weight of residue

4.3.2 Iron - Major Defect - Defect Code 03001.-The iron content shall be determined on the residue from the silica determination.

4.3.2.1 Solutions.-Prepare the following solutions:

o-phenanthroline - 0.25 percent solution of the monohydrate in water
sodium acetate - 0.2M and 2M
hydroxylammonium chloride - 10 percent in water

4.3.2.2 Procedure.-Dissolve the residue in 10 ml of 0.2M sodium acetate solution and add 5 ml of the 10 percent hydroxylammonium chloride solution. Adjust the pH to 3 to 6 with sodium acetate. Transfer to a 100 ml volumetric flask and dilute to 100 ml. To a 25 ml aliquot add 4 ml of o-phananthroline solution, dilute to 50 ml and mix.

NOTE: If the iron content is high, it may be necessary to use an aliquot containing less iron.

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Compare the intensity of the color produced with standards, similarly prepared, in any convenient way. If a colorimeter is employed, use a filter showing maximum transmission at 480-520mμ; for a spectrophotometer, use a wavelength of 510 mμ.

4.3.3 pH - Major Defect - Defect Code 04001.-Mix five (5) grams of material in 100 ml of distilled water for five minutes. Determine the pH of the water, using any standard pH meter standardized using a buffer solution of pH 4.0.

4.3.4 Density - Major Defect - Defect Code 05001.-Weigh 50 ± 0.1 grams of material. Pour slowly through a wide tube funnel into a 100 ml graduated cylinder. Hold the cylinder vertically one (1) inch above a book or thick pad of paper, then drop it three (3) times to the book. Read the volume occupied by the material to the nearest ml.

4.3.5 Water vapor adsorption capacity - Major Defect - Defect Code 06001.-

4.3.5.1 Apparatus.-The general arrangement of the apparatus shall be as shown in Figure 1. The fitting comprising the inlet, outlet and petticoat bubbler for each saturator bottle shall have a ground glass stopper to fit the bottle. The saturator bottle and the bubbler shall be so designed that with one liter of solution in the bottle, the upper holes of the bubbler shall be not more than 3, nor less than 1, inches below the surface of the liquid. Each of the six saturator bottles shall be filled with one liter of identical sulfuric acid solutions to give the required nominal relative humidity at 25°C. Table II shows the solutions to be used for this purpose. The sample-containing bottle shall be a Nesbitt, Fleming or other standard form of adsorption bulb. The bulb shall be provided with a suitable closure to prevent diffusion. The room in which the tests are being run should be controlled to 25 ± 1°C. It is essential that the temperature variation within the saturator bottles, the inlet air, and the sample-containing bottle during any one run shall vary not more than plus or minus 0.5°C. A desiccant trap filled with the silica gel under test shall be placed in front of the saturator bottles to pick up any adsorptive impurities in the inlet air.

4.3.5.2 Procedure.-A 6- to 10-gram sample of the silica gel shall be weighed to the nearest milligram in the tared adsorption bulb. Extreme care shall be exercised to insure that the material is exposed to the air for a minimum time. The adsorption bulb shall then be connected to the apparatus and the air flow adjusted to 4 ± 0.5 liters per minute with the bypass closed. If the manometer indicates the pressure in the last bottle to be greater than 1.0 inch of mercury, the bleed-off line should then be cracked open until

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the manometer indicates less than 1 inch and that point maintained for the balance of the run. When using the bypass, an occasional check shall be made of the amount of air passing through the adsorption bulb by diverting the air from the flowmeter to atmosphere through the "tee" connection. The flowmeter reading will then indicate the actual air flow through the adsorption bulb. Periodically the adsorption bulb shall be removed from the train, the stop-cocks closed, and the bulb weighed. This process shall be repeated until two successive weighings, approximately 1 hour apart, show a weight variation not exceeding 0.1 percent; however, the test shall be considered complete if at any time the minimum specification values are attained. It will generally be found that the weight will rise to a maximum and then decrease slightly before approaching a constant value. The moisture adsorption capacity shall be calculated from the following equation:

$$\text{Moisture adsorption capacity, percent} = 100 \times \frac{Wf - Wo}{Wo - We}$$

Where:

Wf = Final weight of adsorption bulb
Wo = Original weight of bulb and sample
We = Weight of empty bulb

Table II. Relative humidity solutions[1]

Solution	Percent by Weight	RH Percent at 25°C	Specific Gravity[2] 20°/4°C	Specific Gravity 77°/60°C
H_2SO_4	64.61	10	1.5499	1.5462
H_2SO_4	58.21	20	1.4789	1.4754
H_2SO_4	47.85	40	1.3739	1.3715
H_2SO_4	38.36	60	1.2889	1.2859
H_2SO_4	26.23	80	1.1894	1.1853

[1] This table has been computed on the basis of values for the specific gravity found in the International Critical Tables. The nominal relative humidities represent values that would be obtained in a closed static system. The actual effluent relative humidities generally will be slightly below these nominal values.

[2] Specific gravity measurements should be made at either indicated temperature using a precision hydrometer with a smallest subdivision of 0.0005. The specific gravity of the sulphuric acid solution should be checked prior to each determination since the acid-water ratio will change during the course of determination.

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4.3.6 Granulation - Major Defect - Defect Code 07001- Conformance of the material to the granulation requirements of Table I shall be determined using sieves conforming to RR-S-366. Place 100 grams ± 0.1 gram of the sample on the top sieve of the specified nest of seives, properly superimposed and assembled with a bottom pan. Cover the assembly and shake for five (5) minutes by means of a mechanically operated sieve shaker which imparts to the sieves a rotary action and a tapping action. The portions retained or passed by the various sieves shall be weighed and the results calculated to a percentage basis.

5. PREPARATION FOR DELIVERY

5.1 Packaging.-

5.1.1 Level C.-The material shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Marking.-In addition to any special marking required by the contract or purchase order, the containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.-The material covered by this specification is intended for use in Aerial Mine.

6.2 Ordering data.-Procurement documents shall specify the title, number and date of this specification.

6.3 Inspection code numbers.-The five-digit code numbers assigned to the inspection herein are to facilitate future data collection and analysis by the Government.

6.4 One type of silica gel which has been found satisfactory for the intended purpose is a product manufactured by Davidson Division, W. R. Grace & Co., Baltimore, Md., designated as "Davidson No. 5". Materials from other sources shall be evaluated by Picatinny Arsenal, Dover, New Jersey 07801 (SMUPA-ND8) before being approved as an equivalent.

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Preparing activity
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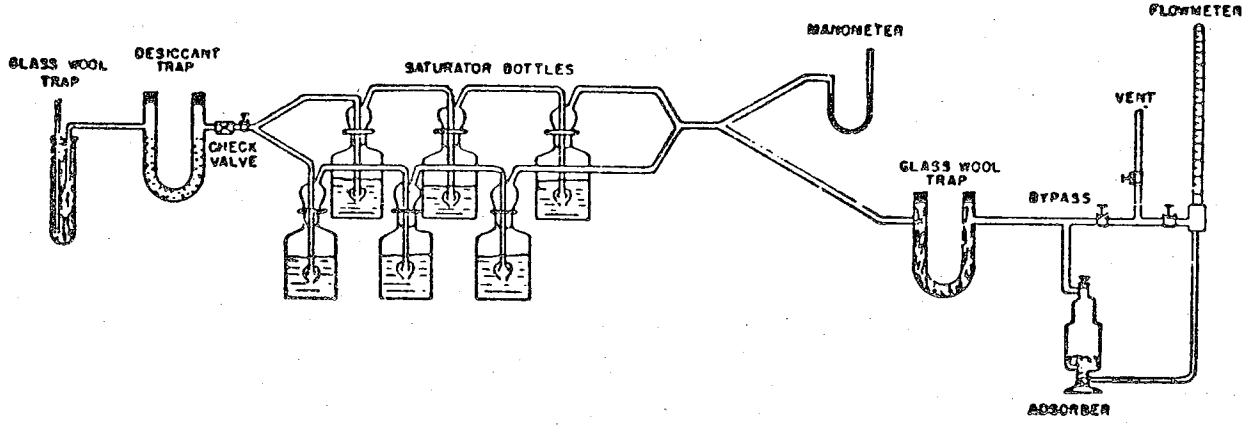


FIGURE 1. *Apparatus for water vapor adsorption capacity test.*

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